
Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising via Explicit Predicate Maps and Stable Fuzzy Allocation

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ABSTRACT ABSTRACT PLACEHOLDER. Denoisers for optical coherence tomography are trained to reduce a pixel error, so they can raise fidelity while lowering the properties a clinician reads, namely layer contrast, boundary definition and the visibility of weak structures. This paper treats the repair of those properties as a separate problem. A frozen denoiser is kept unchanged and a small policy is attached to it. The policy measures six named clinical properties in closed form, converts them into a per pixel decision through five explicit rules, and limits how far the image may move. We prove that the decision layer responds to a change in evidence by at most an exact constant below 0.75, and we measure the realised response against that constant on real images. We give three conditions under which the correction cannot lower contrast to noise ratio, and report how often they hold. The policy adds 168 630 trainable parameters to a backbone of about seven million. Across four backbone families and three datasets from three sources it improves the clinical measures at a small and bounded cost in fidelity, and it transfers to two other scanners with no adaptation of any kind. **RESULTS** PLACEHOLDER.

INDEX TERMS Optical coherence tomography, image denoising, neuro-symbolic reasoning, fuzzy logic, interpretable image restoration, modular post-correction, clinical quality assessment.

I. Introduction

OPTICAL coherence tomography (OCT) is the standard non invasive way to image the retina [1]. Every OCT image carries speckle, a multiplicative disturbance that arises from coherent light scattering inside tissue [2]. Speckle blurs the border between retinal layers, lowers the contrast between tissue and background, and makes automatic layer segmentation less reliable [3].

Deep denoisers remove much of that disturbance. They are almost always trained to minimise the squared error against a clean reference, so they are rewarded for average pixel accuracy and nothing else. A denoiser can therefore raise the peak signal to noise ratio (PSNR) of a scan while making the scan less useful to a clinician. Three things commonly get worse. Layer boundaries become smooth and lose definition. The contrast between tissue and the dark vitreous falls. Thin structures with

weak signal fade into the background. None of these losses is visible in a PSNR number, and none of them is corrected by training the same objective harder.

A. Motivation

The gap we address is not a missing denoiser. Strong denoisers already exist. The gap is that nothing sits between the denoiser and the clinical reader to state, in explicit terms, what a good OCT image should look like and to repair the specific ways in which a denoised image falls short. Building that missing part as another neural block would reproduce the original problem, because the reason for every change would again be hidden inside learned weights. A clinician, a regulator, or a reviewer who sees a modified scan needs to know which property was judged deficient, where in the image that judgement

was made, and how large a change was allowed as a result.

This paper therefore treats post correction as its own problem. We keep the denoiser frozen and we attach a small policy that decides where the image should be changed and by how much. The policy is written down as named image properties, explicit logical rules over those properties, and explicit limits on the size of any change. Nothing in the decision path is an unnamed attention map.

B. New ideas in this paper

We do not claim that edge filters, confidence maps, or fuzzy operators are new on their own. The new ideas are the following five.

- 1) Post correction written as a constrained optimisation problem. Section III states the objective, the decision variables, the constraints, and the solver. The clinical goal becomes a term in an objective rather than an informal aim, and the safety requirements become constraints that are checked on every image at test time.
- 2) Six clinical properties given in closed form. Section D defines six ground truth free predicates. Each one returns a score for the whole image and a map that says where in the image the property is unsatisfied. Both are written as formulas, so the input to the rule layer can be recomputed by anyone from the released code.
- 3) A fuzzy allocation rule with a proved and measured sensitivity bound. Section E builds the allocation from Lukasiewicz connectives. Section V defines what stability means here, proves a Lipschitz bound for the allocation, and Section I measures the true sensitivity on real images and compares it with the bound.
- 4) A safety layer that decides how much of a proposal to keep. Section G states three constraints at three different scales and shows how the number of satisfied constraints sets the amount of the proposed change that is applied. A margin condition under which the correction cannot lower the contrast to noise ratio is proved and then tested numerically.
- 5) Transfer evidence that separates two different claims. Section C reports two protocols that earlier work often mixes together. In the first the policy trained on one scanner is applied to another scanner with no further training. In the second it is allowed a small amount of adaptation on images from the new scanner. We report both so the reader can see what transfers for free and what has to be paid for.

C. How the paper is organised

Section II reviews related work. Section III states the notation and the optimisation problem. Section IV describes the method from the top down, one stage at a time. Section V gives the formal results. Section VI gives the training algorithm and every constant it uses. Section VII reports the tests. Section VIII discusses what the results mean and where the method fails, and Section IX concludes.

II. Related Work

A. Denoising of OCT images

Learned denoisers for OCT fall into three families. Convolutional networks such as DnCNN [4] and DRUNET [3] learn a residual between the noisy and the clean image. Attention based designs such as NAFNet [5] and KBNNet [11] reweight features before combining them. Transformer designs such as SwinIR [6] use windowed self attention. Classical estimators remain in use, including non local means [12] and wavelet shrinkage [13], and self supervised training removes the need for clean targets [7]. Methods built for OCT address the scarcity of paired data directly, through triplet cross fusion [14] and through sparse representation [15], [16].

All of these methods share one property. They are trained to reduce a pixel error, so the quantity they optimise is not the quantity a clinician reads. Work on automated quality assessment for OCT [17], [23], [24] measures clinical quality but does not act on it. The present paper closes that loop by turning measured quality into a correction that is applied under stated limits.

B. Confidence and uncertainty in medical imaging

Dropout used as an approximate Bayesian posterior [18], task weighting driven by learned variance [19], and deep ensembles [20] are the standard ways to attach a confidence value to a prediction. In OCT, such values have been used to flag unreliable layer segmentation [3] and poor scan quality [17]. In almost all of this work the confidence value is reported and then left alone. It marks a region as doubtful without changing what happens there. The method in this paper uses a confidence value as one input to a correction decision, and Section H tests whether that value behaves like a real error predictor rather than a free gating map.

C. Symbolic reasoning and expert networks joined with neural networks

Neuro symbolic computing [21] couples a learned model with explicit rules. Differentiable fuzzy logic makes that coupling trainable, because the connectives become smooth functions of continuous truth values [10]. In medical imaging the pattern has been used to generate spinal reports from images by pairing a generative network with

a symbolic graph of domain knowledge [33], to predict patient deterioration by placing a probabilistic logic program on top of a convolutional feature extractor [34], and to interpret corneal topography through a knowledge graph attached to a learned feature model [40]. Fuzzy membership functions have also been embedded inside networks that process OCT directly, where a rough fuzzy discretisation guides layer segmentation [35], and pixel level fuzzy rules have replaced a learned attention block so that the resulting map can be read as a logical statement [36].

A parallel line of work builds an image model as a set of cooperating experts. Steered mixtures of experts model an image by local regressors combined through an explicit steering term [37], and the same idea has been applied to OCT speckle removal [38]. Recent restoration networks route an input through experts of different capacity according to how hard the input is [39]. Outside imaging, evolving fuzzy models have been used to identify actuator dynamics from data [30], learned models have been combined with a physical forward operator in ultrasound computed tomography [31], and the sensitivity of a differential model to its initial conditions has been studied as a source of uncertainty in simulation [32]. These works show the same underlying principle that we rely on, which is that a learned component becomes easier to trust when part of the decision is written in a language a person can read.

What is still missing in this literature is a rule layer that acts on the output of a strong denoiser, that names the clinical properties it is protecting, and that limits how far it may move the image. That is the object studied here.

III. Notation and Problem Statement

This section fixes the notation used in the rest of the paper and states the problem that the method solves. Every symbol is defined here before it is used anywhere else.

A. Notation

Images are real valued arrays of height H and width W with values in $[0, 1]$. We write $\mathbf{y}$ for the noisy B scan that comes off the scanner and $\mathbf{x}$ for the clean reference obtained by averaging many repeated scans of the same location. A pixel index is written p , so y_p is one pixel of $\mathbf{y}$.

Two products appear. The symbol $\odot$ is the Hadamard product, which multiplies two arrays of the same shape entry by entry, so $(\mathbf{a} \odot \mathbf{b})_p = a_p b_p$. Ordinary multiplication of two numbers is written by juxtaposition with no symbol between the factors. Table 1 lists every remaining symbol in the order in which the paper introduces it.

TABLE 1. Symbols used in the paper, in order of first use.

Symbol	Meaning
$\mathbf{y}$	Noisy input B scan
$\mathbf{x}$	Clean reference B scan
f_θ	Frozen denoising backbone with parameters θ
$\mathbf{b}$	Backbone output, $\mathbf{b} = f_\theta(\mathbf{y})$
ϕ	Trainable parameters of the wrapper
$\mathbf{z}$	Backbone feature map read by the confidence head
$\mathbf{c}$	Confidence map with values in $[0, 1]$
$\mathbf{u}$	Uncertainty map, $\mathbf{u} = \mathbf{1} - \mathbf{c}$
s_i	Score of clinical property i , a number in $[0, 1]$
$\mathbf{m}_i$	Failure map of clinical property i , values in $[0, 1]$
τ_i	Decision threshold for property i
f_i	Soft failure signal derived from s_i and τ_i
Δ	Raw multiplicative gain proposed by the gain network
λ	Per pixel strength map with values in $[0, 1]$
$\mathbf{a}$	Allocation map produced by the rule layer
α	Gain after allocation, strength and clamping
$\mathbf{g}$	Intensity gate that suppresses change in the background
δ_e	Additive edge correction
$\mathbf{q}$	Candidate image before the safety decision
n	Number of satisfied safety constraints, an integer in $\{0, 1, 2, 3\}$
w	Blend weight chosen from n
$\hat{\mathbf{x}}$	Final output of the wrapper
$\mathcal{T}, \mathcal{B}$	Sets of tissue pixels and background pixels
$\mu_{\mathcal{T}}, \mu_{\mathcal{B}}$	Mean value over tissue and over background
$\sigma_{\mathcal{B}}$	Standard deviation over background

B. The optimisation problem

The denoiser is given and is not changed. The task is to produce an image that beats $\mathbf{b} = f_\theta(\mathbf{y})$ on named clinical measures, stays close to it everywhere, and can explain every change it makes. We now state the problem in the form that the training procedure actually solves. This makes precise what is being optimised, over what, and subject to what.

Decision variables. The only free quantities are the wrapper parameters ϕ . These are the confidence head, the gain network, the edge network, the strength predictor and the rule layer. The backbone parameters θ are held fixed throughout.

Objective. Let $\mathcal{D}$ be a set of paired scans. Let $\hat{\mathbf{x}}(\phi)$ be the output of the wrapper. We minimise the expected value of a weighted sum of five terms,

$$\min_{\phi} \mathbb{E}_{(\mathbf{y}, \mathbf{x}) \sim \mathcal{D}} \left[\mathcal{L}_{\text{fid}} + w_c \mathcal{L}_{\text{clin}} + w_p \mathcal{L}_{\text{pred}} + w_o \mathcal{L}_{\text{coop}} + w_s \mathcal{L}_{\text{edge}} \right], \quad (1)$$

where $\mathcal{L}_{\text{fid}}$ keeps the output near the clean reference, $\mathcal{L}_{\text{clin}}$ rewards improvement of the clinical measures over the backbone, $\mathcal{L}_{\text{pred}}$ calibrates the property scores against their thresholds, $\mathcal{L}_{\text{coop}}$ ties the amount of correction to the estimated uncertainty, and $\mathcal{L}_{\text{edge}}$ supervises the additive edge branch. Section VI gives each term in full and lists the weights.

Constraints. Three requirements must hold for the output on every image. They are stated formally in Section G and summarised here. The aggregate clinical quality must not fall by more than a fixed tolerance. No single clinical property may fall by more than a fixed tolerance while another rises. The relative change from the backbone output must stay below a fixed bound.

How the problem is solved. The objective is differentiable in ϕ , including the rule layer, because every logical connective is replaced by a smooth piecewise linear surrogate. We therefore solve the problem by mini batch stochastic gradient descent with the AdamW optimiser [27]. The three constraints are handled in two ways. During training they enter the objective as penalties, so the optimiser is pushed toward feasible solutions. At test time they are evaluated exactly on each image, and the number of satisfied constraints decides how much of the proposed change is applied. That second mechanism is what makes the constraints binding rather than advisory, and it is described in Section G.

IV. Method

A. Overview

Fig. 1 shows the whole method. It runs in six stages and each stage answers one question.

- 1) The frozen backbone produces a denoised image, and a small head estimates how confident the backbone is at each pixel. This answers where the backbone is unsure.
- 2) Two small networks propose changes. One scales the image, the other adds detail back to boundaries. This answers what change is available.
- 3) Six named clinical properties are measured on the backbone output. Each returns a score for the image and a map of where it is unsatisfied. This answers what is wrong and where.
- 4) A rule layer turns those measurements into a per pixel allocation, which says how much of the proposal to admit at each location. This answers where the change should be applied.

- 5) The admitted change is bounded, gated away from the background, smoothed and combined with the edge term. This answers how the change is applied.
- 6) Three constraints are checked on the result, and the number that pass decides how much of the candidate replaces the backbone output. This answers whether the change is allowed at all.

The rest of this section takes the six stages in order.

B. Stage one, the backbone and the confidence map

The backbone is frozen after training and gives $\mathbf{b} = f_{\theta}(\mathbf{y})$. A two layer network h_{ϕ} reads a feature tensor $\mathbf{z}$ from inside it and produces the confidence map

$$\mathbf{c} = \sigma \left(\frac{h_{\phi}(\mathbf{z}) + \beta_0}{\tau_0} \right), \quad (2)$$

where σ is the logistic function, $\tau_0 > 0$ is a learnable temperature and β_0 a learnable offset. The uncertainty map is $\mathbf{u} = \mathbf{1} - \mathbf{c}$. A small temperature gives a nearly binary map and a large one spreads the decision out.

Only this head depends on the backbone, because architectures expose feature tensors with different channel counts. Table 2 lists the tensor used for each. Everything after this stage is shared, which is the precise sense in which the method is modular rather than architecture free. Section K reports what adapting this head costs.

C. Stage two, the two correctors

Two changes are proposed, and they are deliberately of different kinds.

The gain network proposes a multiplicative change. It runs at quarter resolution and emits a brightness channel passed through a rectifier, so it can only brighten, and a contrast channel passed through a hyperbolic tangent, so it can go either way. Their sum is scaled by $\alpha_{\text{max}} = 0.15$ to give the raw gain Δ . A multiplicative form suits speckle, which is itself multiplicative [2], so bright tissue receives a larger absolute change than dark background for the same relative adjustment.

The edge network proposes an additive change δ_e from four channels, namely the intensity normalised residual, the edge magnitude of the backbone output, the backbone output, and an unsharp mask. Its output is bounded to ± 0.08 and scaled by a learned factor. An additive form suits boundaries, which should be sharpened by the same amount in a bright layer as in a dark one.

A third small network predicts a per pixel strength map λ with values in $[0, 1]$ from the failure maps of Stage three. It modulates how much of the raw gain survives at each pixel.

D. Stage three, the six clinical properties

The six properties are the interface between clinical language and the arithmetic of the method. Each one is

chosen because it names a way a denoised OCT image is known to fail, and each is written as a formula so it can be recomputed by anyone.

Table 3 gives, for every property, the clinical reason it exists, the score $s_i \in [0, 1]$ that summarises the whole image, the map $\mathbf{m}_i \in [0, 1]^{H \times W}$ that localises the failure, and the threshold τ_i that separates satisfied from unsatisfied. A larger score means the property is better satisfied. A larger value of the map means stronger local evidence that it is not.

None of these quantities needs a clean reference. They are computed from the noisy input and the backbone output only, which is why they remain available at test time on a scanner the method has never seen. Fig. 2 shows the six maps for one image.

E. Stage four, the rule layer

The rule layer decides where the gain proposal should act. It does so through five named rules, each of which casts a signed vote.

Each property score is first turned into a soft failure signal by comparing it with its threshold,

$$f_i = \sigma(\kappa(\tau_i - s_i)), \quad \kappa = 10, \quad (3)$$

so f_i is near one when property i is failing and near zero when it is satisfied. The constant κ sets how sharp that transition is. The same soft comparison is used to test the confidence map and the strength of a proposal against their own thresholds. Conjunction uses the Lukasiewicz form $\text{AND}(a, b) = \max(0, a + b - 1)$, which is piecewise linear and has a bounded slope, which is what makes the bound in Section V tight.

The five rules are the following. Rule one fires when the backbone is confident and the proposal is weak, and it votes against changing the image. Rule two fires when the backbone is unsure and the proposal is strong, and it votes for changing it. Rule three fires when a relevant property is failing, and it votes for changing it. Rule four fires when confidence and proposal strength are both middling, and it votes weakly for changing it. Rule five fires when several correctors want the same pixel, and it votes against changing it. Write r_1 to r_5 for the resulting truth values.

The allocation at each pixel is the bounded weighted vote

$$a = \underbrace{B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta)}_{\text{base, no rule firing}} - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 + C_4 w_4 r_4 - C_5 w_5 r_5, \quad (4)$$

where β is a learnable base logit, $w_k = \sigma(\theta_k) \in (0, 1)$ are learnable rule weights and the budgets are $B_{\min} = 0.25$, $B_{\max} = 0.50$, $C_1 = 0.15$, $C_2 = 0.25$, $C_3 = 0.15$, $C_4 = 0.10$ and $C_5 = 0.10$.

Those budgets are not arbitrary. They are chosen so that $B_{\max} + C_2 + C_3 + C_4 = 1$ and $B_{\min} - C_1 - C_5 = 0$.

Since every truth value lies in $[0, 1]$ and every weight lies in $(0, 1)$, the value of Equation (4) already lies in $[0, 1]$ for every possible input and every possible parameter value. No clipping is ever needed. This matters for two reasons. A clipped allocation would stop the gradient, so the rules could never be trained, and a bound that is enforced by clipping would make the stability result in Section V vacuous.

F. Stage five, applying the change

The admitted gain is the raw proposal modulated by the allocation and the strength map, then bounded,

$$\alpha = \text{clamp}(\Delta \odot \mathbf{a} \odot \lambda, -0.20, 0.20). \quad (5)$$

A gate suppresses any change in the dark background,

$$\mathbf{g} = \sigma(20(\mathbf{b} - q_{0.15})), \quad (6)$$

where $q_{0.15}$ is the fifteenth percentile of the backbone output of that image. Using a percentile rather than a fixed level makes the gate work across scanners with different brightness.

The gated gain is applied multiplicatively, and the brightening part is smoothed before it is added so that it cannot create a new high frequency edge,

$$\mathbf{v} = (\text{smooth}(\text{ReLU}(\alpha) \odot \mathbf{b}) + (\alpha - \text{ReLU}(\alpha)) \odot \mathbf{b}) \odot \mathbf{g}, \quad (7)$$

where smooth is a three by three mean filter. The darkening part is left unsmoothed because it has zero mean and cancels itself. In the background, defined by the complement of the gate eroded by two pixels, a masked three by three mean filter is blended in at half strength, which lowers the background standard deviation without letting blur cross into tissue. Finally the edge proposal is added and the result is clipped to the valid range,

$$\mathbf{q} = \text{clamp}(\mathbf{b} + \mathbf{v} + \delta_e \odot \mathbf{g}, 0, 1). \quad (8)$$

G. Stage six, the safety decision

The candidate $\mathbf{q}$ is not returned directly. Three constraints are checked on it.

Aggregate quality must not fall. Define the energy

$$E(\mathbf{v}) = 1 - \frac{1}{6} \sum_{i=1}^6 s_i(\mathbf{v}) + 0.1 \sum_{i=1}^6 \max(0, 0.5 - s_i(\mathbf{v})), \quad (9)$$

which is small when the image is good. The first constraint requires $E(\mathbf{q}) \leq E(\mathbf{b}) + 0.10$. The second term of Equation (9) adds a penalty for any single property that falls below one half, so a good average cannot hide one badly broken property.

No property may be sacrificed for another. The second constraint requires that at least one property improves by more than 0.01 and that no property falls by more than 0.10.

The image must not move far. The third constraint requires $\|\mathbf{q} - \mathbf{b}\|_2 / \|\mathbf{b}\|_2 < 0.2$.

TABLE 2. Feature tensor read by the confidence head for each backbone. All heads share the same two layer design and differ only in the number of input channels.

Backbone	Feature tensor	Channels
NAFNet	First encoder stage	40
KBNet	First encoder stage	32
DnCNN	First convolution	228
SwinIR	Shallow feature convolution	138

TABLE 3. The six clinical properties. Each one names a way a denoised OCT image is known to fail, returns a score for the whole image and a map of where the property is unsatisfied, and is compared against a threshold. The closed form of every score and every map is given in the supplementary file and in the released code. Here e is the Sobel edge magnitude of the backbone output, σ_k is the local standard deviation over a window of size k , and cv is the local coefficient of variation of the log residual.

P_i	Property	Where the map is large	τ_i
P_1	edge continuity	away from detected edges, where a boundary should continue but does not	0.60
P_2	local contrast	where σ_9 falls below the image mean, and in nearly flat tissue	0.50
P_3	flat smoothness	inside flat regions that still carry residual variation	0.75
P_4	structural coherence	where the edge map is itself rough, a sign of broken structure	0.50
P_5	speckle conformity	where cv departs from the value expected for OCT speckle	0.30
P_6	layer visibility	where the local variance of the edge map is high across depth	0.70

Let $n \in \{0, 1, 2, 3\}$ be the number of constraints that hold. The output is

$$\hat{\mathbf{x}} = (1 - w) \mathbf{b} + w \mathbf{q}, \quad w = \begin{cases} 1, & n \geq 2, \\ 0.05 + 0.10 \frac{n}{3}, & n < 2. \end{cases} \quad (10)$$

When at least two constraints hold the candidate is accepted in full. When fewer hold the output falls back almost entirely to the backbone, retaining between five and eight percent of the proposal so that the decision remains visible in the output rather than silently discarded. This rule is evaluated on every image at test time, and Section I reports how often each constraint is the one that fails.

V. Stability and Guarantees

This section states what the method guarantees. We first define the notion of stability that is used, because the word carries several meanings, then prove that the rule layer has it, and then give the condition under which the correction cannot reduce contrast. Section I tests both results numerically on real images.

A. What stability means here

This is not stability in the sense of Lyapunov and we claim no fixed point. The quantity that must not behave erratically is the decision the rule layer makes, which is a function of the five rule truth values. Those values are exactly what shift when the backbone or the scanner changes.

Definition 1 (Stability of the allocation). Let $\mathbf{r} = (r_1, \dots, r_5) \in [0, 1]^5$ be the vector of rule truth values at one pixel and let $a(\mathbf{r})$ be the allocation of Equation (4) at that pixel with the parameters held fixed. The allocation is said to be stable with constant L if

$$|a(\mathbf{r}) - a(\mathbf{r}')| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty \quad \text{for all } \mathbf{r}, \mathbf{r}' \in [0, 1]^5. \quad (11)$$

In words, a small change in the evidence produces a proportionally small change in the decision, and L is the worst case ratio between the two. A method without such a bound could swing from applying no correction to applying all of it because one property score moved slightly, which is precisely the behaviour a clinical user cannot audit.

B. The rule layer is stable, with an exact constant

Theorem 1 (Allocation stability). The allocation of Equation (4) satisfies Definition 1 with

$$L = C_1 w_1 + C_2 w_2 + C_3 w_3 + C_4 w_4 + C_5 w_5, \quad (12)$$

where $w_k = \sigma(\theta_k)$ are the learned rule weights. The constant is attained, so it is the exact Lipschitz constant and not merely an upper bound. Moreover $L < C_1 + C_2 + C_3 + C_4 + C_5 = 0.75$ for every possible setting of the parameters.

Proof:

Write $s_k = -1$ for $k \in \{1, 5\}$ and $s_k = +1$ otherwise, so that Equation (4) reads $a(\mathbf{r}) = b + \sum_k s_k C_k w_k r_k$ with b independent of $\mathbf{r}$. Subtracting two such expressions gives

$$a(\mathbf{r}) - a(\mathbf{r}') = \sum_k s_k C_k w_k (r_k - r'_k).$$

Taking absolute values and applying the triangle inequality,

$$|a(\mathbf{r}) - a(\mathbf{r}')| \leq \sum_k C_k w_k |r_k - r'_k| \leq \left(\sum_k C_k w_k \right) \|\mathbf{r} - \mathbf{r}'\|_\infty,$$

which is Equation (11) with L as claimed. The bound is attained by taking $r_k - r'_k = s_k t$ for any t small enough

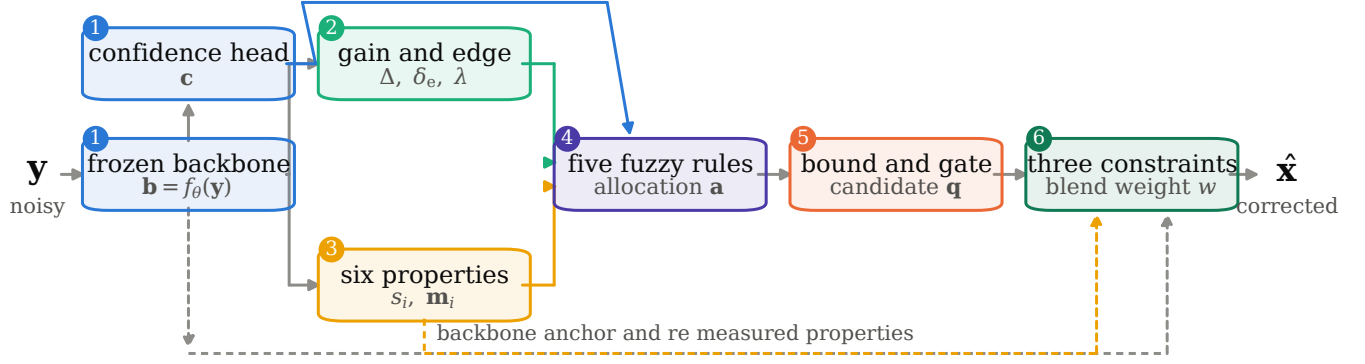


FIGURE 1. The six stages of the method. A frozen backbone produces the denoised image and a small head estimates confidence. Two networks propose a multiplicative change and an additive edge change. Six clinical properties are measured and produce a score and a failure map each. Five rules turn those signals into a per pixel allocation. The admitted change is bounded, gated away from the background and combined. Three constraints are then checked, and the number that pass decides how much of the candidate replaces the backbone output.

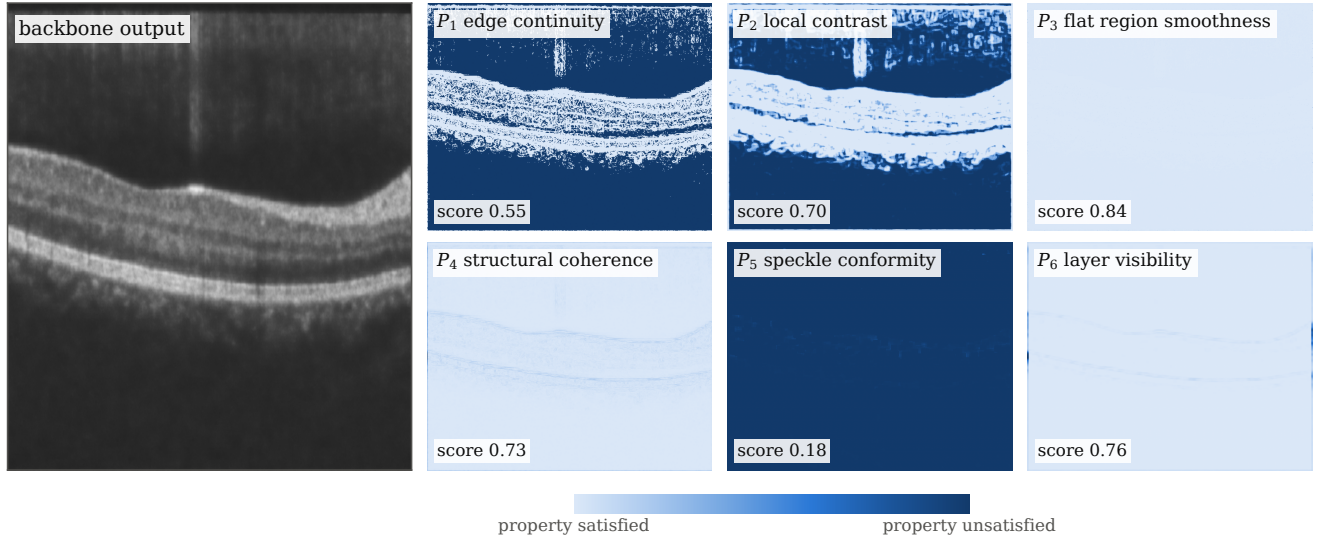


FIGURE 2. The six clinical properties on one backbone output. Each panel shows where that property is locally unsatisfied, on a shared scale, with the image level score in the corner. Edge continuity, local contrast and speckle conformity carry most of the evidence on this image, while the other three are close to satisfied everywhere.

to keep both vectors in the unit cube, since then every term of the sum has the same sign. Because each C_k is positive and each $w_k = \sigma(\theta_k)$ lies strictly in $(0, 1)$, the sum is strictly less than $\sum_k C_k = 0.75$. ■

Two properties do the work. The allocation is affine in the rule truth values, which makes the constant exact rather than conservative. The budgets satisfy $B_{\max} + C_2 + C_3 + C_4 = 1$ and $B_{\min} - C_1 - C_5 = 0$, so no clipping is ever applied. Had a clip been active the bound would still hold, but the gradient would vanish wherever it was tight and the rule weights could not be trained at all.

Corollary 1 (Behaviour under a change of backbone). Suppose replacing one backbone by another changes the rule truth values at a pixel by at most ϵ in the maximum norm, with the parameters unchanged. Then

the allocation at that pixel changes by at most $L\epsilon$, with L from Equation (12).

This is a statement about the routing rule alone. It says the decision layer degrades gracefully when the evidence shifts. It is not a claim that image quality transfers, which is an empirical question answered in Section C.

C. When the correction cannot reduce contrast

Contrast to noise ratio is the measure this method targets most directly. Write $\text{CNR}(\mathbf{v}) = (\mu_{\mathcal{T}}(\mathbf{v}) - \mu_{\mathcal{B}}(\mathbf{v})) / \sigma_{\mathcal{B}}(\mathbf{v})$ for the separation between tissue and background divided by the variability of the background.

Theorem 2 (Contrast margin). Let $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$ be the gain in tissue mean and let $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$ bound the leakage into the background. If

the correction does not raise the background variability, that is $\sigma_B(\mathbf{q}) \leq \sigma_B(\mathbf{b})$, then the output of Equation (10) satisfies

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_T(\mathbf{b}) - \mu_B(\mathbf{b})) + w(\eta_T - \eta_B)}{\sigma_B(\mathbf{b})}. \quad (13)$$

In particular, whenever $\eta_T > \eta_B$ the correction cannot reduce the contrast to noise ratio.

The proof is a convex combination argument for the numerator and Minkowski's inequality for the denominator. It is given in full in the supplementary file.

Three parts of the design work to make the hypotheses of Theorem 2 hold in practice. The gate of Equation (6) keeps η_B small by suppressing change in the dark region. The brightness branch of the gain network cannot be negative, which keeps η_T non negative. The background smoothing lowers $\sigma_B(\mathbf{q})$ rather than raising it. Section I measures all three quantities on every test image and reports how often the hypotheses actually hold.

D. What is not claimed

These results are deliberately narrow. Theorem 1 bounds how much the routing decision can move when the evidence moves, and says nothing about whether the decision is correct. Theorem 2 gives a sufficient condition for contrast not to fall, and says nothing about the other five measures. Neither result promises that the corrected image is closer to the clean reference, and in fact Section B reports that peak signal to noise ratio usually falls slightly. The guarantees are about the behaviour of the policy, not about the quality of the denoiser underneath it.

VI. Training

This section gives the training procedure in full. It states what data is used, what each loss term measures, how the parameters are initialised, how every constant was chosen, and what is left fixed.

A. What data is used, and where a clean image is needed

This point matters for the credibility of the method, so we state it plainly. Training uses paired data. Each training sample is a noisy scan together with a clean reference produced by averaging many repeated acquisitions. The clean reference enters two of the five loss terms, namely the fidelity term and the clinical term, and it also supervises the additive edge branch. Nothing in the method pretends to be unsupervised.

Testing uses no clean image at any point. The six clinical properties of Section D are computed from the noisy input and the backbone output alone. That is what the phrase ground truth free means in this paper. It refers to the inputs of the rule layer at test time, not to the training procedure. The clean reference is used to score the results, as it is in any denoising study, but it never reaches the correction path.

B. The five loss terms

Let $\mathbf{x}$ be the clean reference, $\mathbf{b}$ the backbone output and $\hat{\mathbf{x}}$ the wrapper output. The objective of Equation (1) has the following terms.

Fidelity. This term keeps the output near the reference. It combines the squared error with the structural similarity index [26],

$$\mathcal{L}_{\text{fid}} = \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2 + (1 - \text{SSIM}(\hat{\mathbf{x}}, \mathbf{x})). \quad (14)$$

A dead zone is applied to the peak signal to noise ratio part of this term. No penalty is charged while the drop with respect to the backbone stays below d decibels. The purpose is to buy clinical quality with a small and bounded amount of fidelity rather than to defend fidelity at any price.

Clinical improvement. This term rewards a rise in the clinical measures relative to the backbone. Working with the logarithm of a ratio makes the term scale free, so an image with a low starting contrast is not treated differently from an image with a high one,

$$\mathcal{L}_{\text{clin}} = -\log \frac{\text{CNR}(\hat{\mathbf{x}})}{\text{CNR}(\mathbf{b})} - \log \frac{\text{TCI}(\hat{\mathbf{x}})}{\text{TCI}(\mathbf{b})}. \quad (15)$$

The implementation adds the same construction for the edge preservation index, the equivalent number of looks and the signal to noise ratio, each with its own weight, so that a gain in one measure cannot be bought by a collapse in another.

Property calibration. The rule layer compares each property score with a threshold. If the scores drift away from the thresholds the comparison becomes uninformative. This term pulls the scores toward the operating point,

$$\mathcal{L}_{\text{pred}} = \sum_{i=1}^6 (s_i(\hat{\mathbf{x}}) - \tau_i)^2. \quad (16)$$

Cooperation. The wrapper should act where the backbone is unsure and stay quiet where it is confident. We measure the agreement between the uncertainty map and the map of estimated correction potential with the Pearson correlation ρ , and penalise disagreement,

$$\mathcal{L}_{\text{coop}} = \frac{1}{2} \text{ReLU}(1 - \rho(\mathbf{u}, \boldsymbol{\pi})). \quad (17)$$

The Pearson correlation is used because it is differentiable everywhere, it is invariant to an affine change of either map, and it introduces no window size to tune.

Edge supervision. The additive branch is trained to reproduce the gradient that the backbone lost, measured with the Sobel operator ∇ ,

$$\mathcal{L}_{\text{edge}} = \|\nabla \hat{\mathbf{x}} - \nabla \mathbf{x}\|_2^2. \quad (18)$$

C. Optimisation schedule

The backbone is trained first on the training split with the squared error alone and is then frozen. Every result in this paper uses that frozen backbone without further change.

The wrapper is then trained in two stages. In the first stage only the clinical terms are active, so the correctors learn to move the image in a useful direction. In the second stage the fidelity term is switched on, so the correctors learn how far they may go. The switch happens at the midpoint of the schedule. We use AdamW [27] with decoupled weight decay of 10^{-4} , random crops of 96×96 pixels, and a batch of four crops. Three learning rates are used, a lower one for the corrector networks and higher ones for the strength predictor and the rule layer, because the last two have far fewer parameters and therefore need larger steps to move at all. The procedure is stated step by step in the supplementary file.

D. How the loss weights were chosen

The five terms of Equation (1) carry weights, and those weights decide where the method sits on the trade off between fidelity and clinical quality. We selected them by search, on the validation split only, and we state the rule that made the choice so that it can be checked.

Four settings were prepared in advance, ranging from a conservative one that barely tolerates any loss of fidelity to an aggressive one that allows more than a decibel. Each setting was trained and then scored on the validation images. Among the settings whose loss of peak signal to noise ratio stayed within one decibel and whose six clinical measures were all positive, we took the one with the largest gain in contrast to noise ratio. When no setting satisfied both conditions we took the one with the fewest negative clinical measures, breaking ties by the contrast gain.

The search was repeated separately for each backbone. This matters. A backbone that already produces good contrast needs a gentler correction than one that suppresses it, and forcing a single setting on all four gives the wrong amount of correction to at least one of them. Table 4 reports the setting chosen for each backbone. The test split was not used at any point in this selection, and every number in Section VII comes from a single evaluation of the chosen models on that split.

E. Initialisation and the choice of every constant

Table 4 lists the constants that carry a design decision, where each acts, and how it was chosen. The remaining constants, namely the window sizes, the gate steepness, the crop size and the batch size, are structural and are listed in the supplementary file. We separate three cases. A constant marked measured was set from a property of the data. A constant marked selected was chosen on the validation split and never on the test split. A constant marked fixed is a structural choice that was not tuned. Section E reports how much the results move when the constants of the rule layer are changed, which is the direct test of whether any of them is doing hidden work.

TABLE 4. Every constant in the method, where it acts and how it was chosen. Fixed means a structural choice that was not tuned. Measured means it is computed from the data at run time. Selected means it was chosen on the validation split and never on the test split. Learned means it is a parameter of the model.

Constant	Value	Acts on	Origin
$\alpha_{\max}$	0.15	scale of the raw gain	selected
gain bound	± 0.20	Eq. (5)	selected
gate percentile	0.15	Eq. (6)	measured
κ	10	Eq. (3)	fixed
$B_{\min}, B_{\max}$	0.25, 0.50	base of Eq. (4)	fixed
$C_1 \dots C_5$	0.15, 0.25, 0.15, 0.10, 0.10	rule votes	fixed
$\tau_1 \dots \tau_6$	0.6, 0.5, 0.75, 0.5, 0.3, 0.7	Eq. (3)	learned
$\beta, \theta_1 \dots \theta_5$	learned	Eq. (4)	learned
energy tolerance	0.10	constraint one	selected
Pareto tolerance	0.10	constraint two	selected
change bound	0.2	constraint three	selected
background blend	0.5	background smoothing	selected
learning rates	$2 \times 10^{-4}, 5 \times 10^{-4}, 3 \times 10^{-4}$	training	selected

The networks use the default initialisation of their layers. Two exceptions are deliberate. The bias of the brightness head starts at a small positive value so that the multiplicative branch begins from a mild brightening rather than from zero, which avoids a flat region of the loss at the start of training. The threshold of each property starts at the value reported in the clinical literature for that property and is then free to move.

VII. Tests and Results

A. Setup

1) Data

Three retinal OCT datasets from three different sources are used, listed in Table 5. Paired data is expensive in OCT because a clean reference requires many repeated acquisitions under steady fixation [14], so these are among the largest public paired sets available. PKU37 [14] provides 37 cross sections from a spectral domain scanner with references averaged over 50 frames. We split it by clean image identity into 27 for training, 5 for validation and 5 for testing, which yields 1388, 173 and 173 noisy and clean pairs. Splitting by identity rather than by pair is what prevents a noisy realisation of a training image from appearing in the test set. Duke17 [15] provides 16 paired B scans from a clinical scanner at a different institution. Duke2013 [16] provides 18 volumes with controlled degradation.

2) The transfer protocol, stated exactly

This point was ambiguous in the previous version of this paper and we state it precisely here.

Every model is trained once on the PKU37 training split. The Duke17 and Duke2013 results in Table 7 are produced by applying that model directly to the external images. No parameter is updated on Duke data. No threshold is recalibrated on Duke data. No statistic of the Duke images is used to select anything. The only quantity computed per image is the intensity percentile inside the gate of Equation (6), which is part of the method itself and is computed identically on PKU37.

For completeness we also report the alternative protocol, in which the model is allowed a short adaptation on the other images of the same external dataset before being applied to a held out image. Reporting both separates what transfers without cost from what has to be paid for, and Section C gives the gap between them.

3) Measures

Seven quantities are reported. Peak signal to noise ratio measures fidelity against the clean reference. The other six are clinical and are reported as a percentage change from the backbone output to the corrected output. Contrast to noise ratio measures the separation of tissue from background. Tissue contrast index measures dynamic range inside tissue. Edge preservation index measures agreement of the edge map with the reference. Boundary sharpness measures the gradient across layer boundaries. Equivalent number of looks measures speckle suppression. Signal to noise ratio measures signal strength against the noise floor. Every measure is computed per image and then averaged, never pooled across a batch, because pooling inflates the result.

4) Cost

The wrapper adds 168 630 trainable parameters to a frozen backbone of about seven million, which is an increase of about 2.4 percent, summarised at the foot of Table 5.

B. In distribution results

Table 6 reports the PKU37 test set. TBDRESULTS

C. Transfer to other scanners

Table 7 reports both protocols on Duke17 and Duke2013. TBDRESULTS

D. Contribution of each clinical property

Reviewer feedback asked whether all six properties are needed. We remove one property at a time at test time, which sets its score to the satisfied value and its failure map to zero, and re evaluate the whole test set. Fig. 3 reports the result. TBDRESULTS

E. Sensitivity to the rule constants

The rule layer contains a base level, five rule weights and five budgets. We vary each in turn and re evaluate. Fig. 3 reports the result. We also replace the Lukasiewicz conjunction by the product and by the minimum, which are the two other standard choices, to test whether the specific logic matters. TBDRESULTS

F. Contribution of each component

Table 8 removes one component at a time. TBDRESULTS

G. Comparison at matched complexity

A reviewer asked for a comparison against methods of similar complexity built with the same design approach. We therefore compare against three alternatives that all act on the same frozen backbone output. The first is a plain convolutional post corrector with the same parameter budget, trained with the same objective, with no properties, no rules and no safety layer. The second and third are classical post processing operators, namely unsharp masking and contrast limited adaptive histogram equalisation, tuned on the validation split. The lower block of Table 8 reports the result. TBDRESULTS

H. Does the confidence map predict error

A reviewer observed that nothing in the objective forces the confidence map to behave like a confidence. We test it directly. For every test image we compare the map against the absolute error of the backbone output with respect to the clean reference. The rank correlation, the sparsification error and the expected calibration error after a single affine fit. TBDRESULTS

I. Do the theoretical conditions hold on real data

This subsection connects Section V to measurements, which is what a reviewer asked for. Three things are checked. For the stability result we perturb the rule truth values by a controlled amount and measure the ratio in Equation (11), then compare it with the constant of Equation (12). For the contrast result we measure the tissue gain, the background leakage and the ratio of background standard deviations on every image, and report how often the hypotheses of Theorem 2 hold. For the safety layer we report how often each of the three constraints fails and which one is binding. TBDRESULTS

J. Safety, and whether the correction invents structure

A correction that raises contrast could in principle sharpen a boundary that is not there, or erase a faint one that is. Because PKU37 provides a clean reference we can measure both directly. We report the rate at which an edge appears in the corrected image where the reference has none, the retention of weak structures

TABLE 5. The three datasets. PKU37 is the training domain. The two Duke sets are used only for transfer and no parameter is fitted on them.

Dataset	Scanner	Image size	Images
PKU37 [14]	spectral domain, PKU	640×640	1388 / 173 / 173
Duke17 [15]	spectral domain, Duke	970×520	16
Duke2013 [16]	synthetic degradation	900×450	18

whose contrast in the reference is in the lowest decile, the behaviour inside dark fluid like regions enclosed by tissue, and the worst case per image change across the whole test set. Fig. 3 reports the result. TBDRESULTS

We state one limitation plainly. No clinician read these images. The analysis above is quantitative and uses the averaged reference as the standard of truth. A reader study is the correct next step and is outside the scope of this revision.

K. Cost of moving to a new backbone

Only the confidence head depends on the backbone, and adapting it is additional training, so its cost should be stated rather than assumed away. For each of the four backbones we report the size of the head, the data used to fit it, the number of epochs, the wall clock time and the spread across repeated runs. TBDRESULTS

VIII. Discussion

A. Why fidelity falls, and when that is the right trade

Peak signal to noise ratio usually falls slightly when the wrapper is applied, and the size of the drop differs by backbone. This is not an accident of tuning. It follows from what the two objectives reward. A denoiser trained on squared error is at a local optimum of that error, so any change to its output moves away from that optimum. The wrapper makes changes that raise contrast and boundary definition, and those changes are not free in squared error terms. The question is therefore not how to avoid the drop but how to decide when it is worth accepting.

Two things decide it. A change of a few tenths of a decibel is smaller than the variation between two acquisitions of the same eye, so it does not change what a downstream algorithm can do. And the quantities that improve, namely boundary definition and tissue to background separation, are what a grader actually uses to place a segmentation boundary or judge a fluid pocket.

The backbone that loses most fidelity is the one that had suppressed the most contrast to begin with. A transformer with windowed attention smooths aggressively inside a window, which is efficient for squared error and costly for local contrast. The wrapper restores that contrast, and restoring it necessarily moves the pixels further from the smoothed optimum. The pattern is therefore informative rather than embarrassing. The size

of the correction tracks how much the backbone had given away.

For a deployment that must not lose fidelity at all, the safety layer already provides the control. Tightening the third constraint of Section G reduces the admitted change, and the result moves smoothly toward the backbone output. What the method does not offer is a free lunch in which every measure improves at once, and we prefer to say so.

B. What the formal results buy, and what they do not

Theorem 1 is exact and it is about the code that ran. The constant follows from the trained weights and Section I measures the realised ratio against it. That is a different claim from a bound that holds vacuously because a clip enforces it.

Theorem 2 is a sufficient condition rather than a guarantee of improvement. It says which three quantities have to behave for contrast not to fall, and the method is built to make all three behave. Reporting how often they actually hold, rather than asserting that they must, is the honest form of that claim.

C. Why the correction layer is worth its complexity

A plain network with the same parameter budget can also improve clinical measures, as Section G shows. What it cannot do is say why it changed a pixel. Every change made here can be traced to a named property, a location where that property was measured to fail, a rule that fired, and a safety decision that admitted the result. That trace is the product, and the comparison at matched complexity is what shows the trace is not bought by giving up performance.

D. Limitations

The correction is bounded by construction, so a backbone that has destroyed a structure cannot be made to recover it. The formal results are narrow in the way stated at the end of Section V. The confidence head is the one part that must be refitted for a new backbone, and Section K reports what that costs rather than treating it as free. The clinical measures used here are established proxies, not outcomes, and no clinician read the images, so the safety analysis of Section J is quantitative evidence rather than clinical validation. Finally, the external datasets are small, sixteen and eighteen images, so the transfer results should be read as evidence of behaviour across scanners rather than as a precise estimate of the gain on any one device.

IX. Conclusion

We presented a post correction layer for OCT denoising that is attached to a frozen denoiser rather than replacing it. The method states six clinical properties in closed form, turns them into a per pixel decision

TABLE 6. Results on the PKU37 test set of 173 images. The peak signal to noise ratio is absolute and its change is in decibels. The six clinical measures are reported as the percentage change from the backbone output to the corrected output. TBDRESULTS

Backbone	PSNR (dB)	Δ PSNR (dB)	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL	Δ SNR
NAFNet	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DnCNN	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
SwinIR	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
KBNet	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
mean		TBD	TBD	TBD	TBD	TBD	TBD	TBD

TABLE 7. Transfer to two other scanners. The upper block applies the model trained on PKU37 with no adaptation of any kind. The lower block allows a short adaptation on the other images of the same dataset before testing on the held out image. The gap between the blocks is the price of the adaptation. TBDRESULTS

Protocol	Backbone	Δ PSNR (dB)	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL	Δ SNR
Duke17, no adaptation								
	NAFNet	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	DnCNN	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	SwinIR	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	KBNet	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Duke2013, no adaptation								
	NAFNet	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	DnCNN	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	SwinIR	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	KBNet	TBD	TBD	TBD	TBD	TBD	TBD	TBD
With adaptation, leave one out								
	NAFNet, Duke17	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	NAFNet, Duke2013	TBD	TBD	TBD	TBD	TBD	TBD	TBD

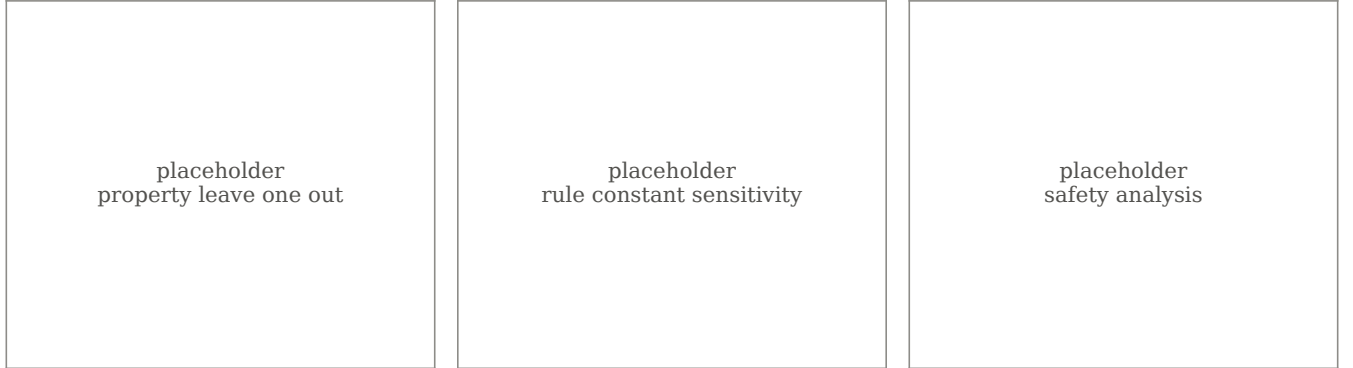


FIGURE 3. Left, the effect of removing one clinical property at a time on the full PKU37 test set. Middle, sensitivity of the result to the constants of the rule layer. Right, the safety analysis. TBDRESULTS

through five named rules, and limits how far the resulting change may move the image. Each of those three parts is written down rather than learned in the dark, so a change to an image can be traced back to the property that motivated it, the place where that property failed, and the constraint that allowed the change to stand.

The decision layer is stable in an exact sense. Its response to a change in the evidence is bounded by a constant that can be computed from the trained weights,

and that constant is attained rather than conservative. Under three stated conditions the correction cannot reduce contrast to noise ratio, and we measure how often those conditions hold rather than assuming them.

Across four backbone families and three datasets from three sources the same correction policy improves the clinical measures at a small and bounded cost in fidelity, with only the confidence head refitted per backbone. The remaining work is clinical rather than technical. The

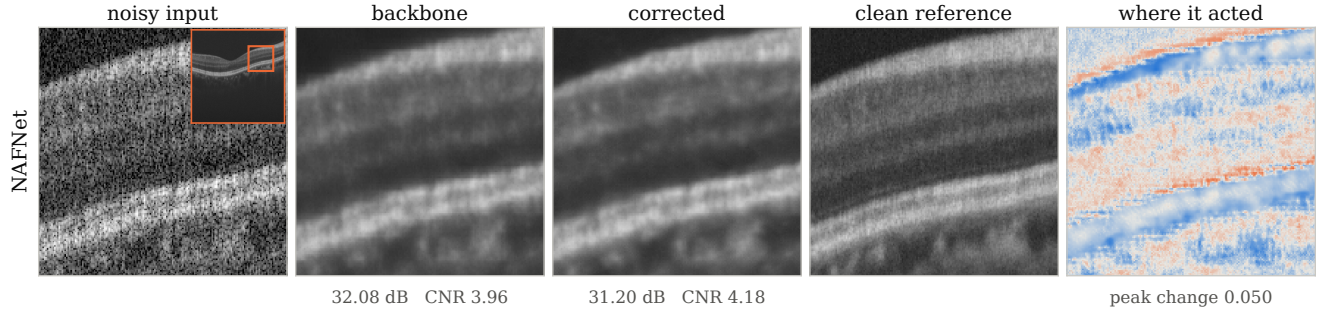


FIGURE 4. Visual comparison on PKU37 for the two extreme cases. Rows are NAFNet, which loses the least fidelity, and SwinIR, which gains the most contrast. Columns are the noisy input, the backbone output, the corrected output and the clean reference. The remaining two backbones are shown in the supplementary file.

TABLE 8. Upper block, removing one component at a time. Lower block, alternatives of similar complexity acting on the same frozen backbone output. TBDRESULTS

Configuration	Δ PSNR	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENI
Full method	TBD	TBD	TBD	TBD	TBD	TBD
No rule layer	TBD	TBD	TBD	TBD	TBD	TBD
No edge branch	TBD	TBD	TBD	TBD	TBD	TBD
No confidence	TBD	TBD	TBD	TBD	TBD	TBD
No background smoothing	TBD	TBD	TBD	TBD	TBD	TBD
Plain corrector, same size	TBD	TBD	TBD	TBD	TBD	TBD
Unsharp masking	TBD	TBD	TBD	TBD	TBD	TBD
Adaptive equalisation	TBD	TBD	TBD	TBD	TBD	TBD

measures used here are proxies, and the natural next step is a reader study that asks whether the changes the method makes are the changes a grader would want.

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